University of Tehran

Faculty of Engineering

Department of Electrical Engineering and Computer Engineering

Introduction To Cognitive Neuroscience

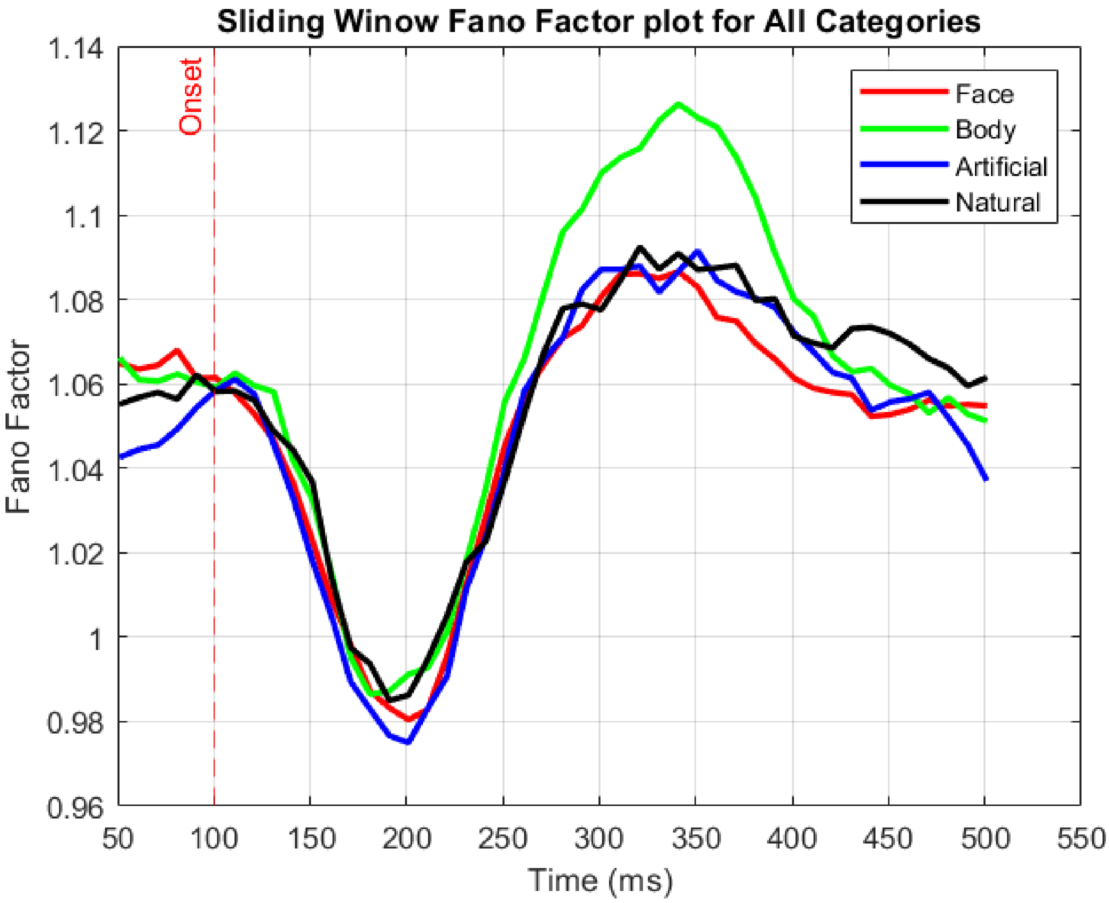
Assignment 2

Phase 1

810102559

May 2025

TABLE OF CONTENTS


Sliding Winow Fano Factor plot for All Categories
Onset
Face
Body
Artificial
Natural
Fano Factor
Time (ms)
1.14
1.12
1.1
1.08
1.06
1.04
1.02
1
0.98
0.96
50
100
150
200
250
300
350
400
450
500
550

Figures Content

Task 1: Spike Sorting From Scratch

1-1. Filtering the Data and Histogram

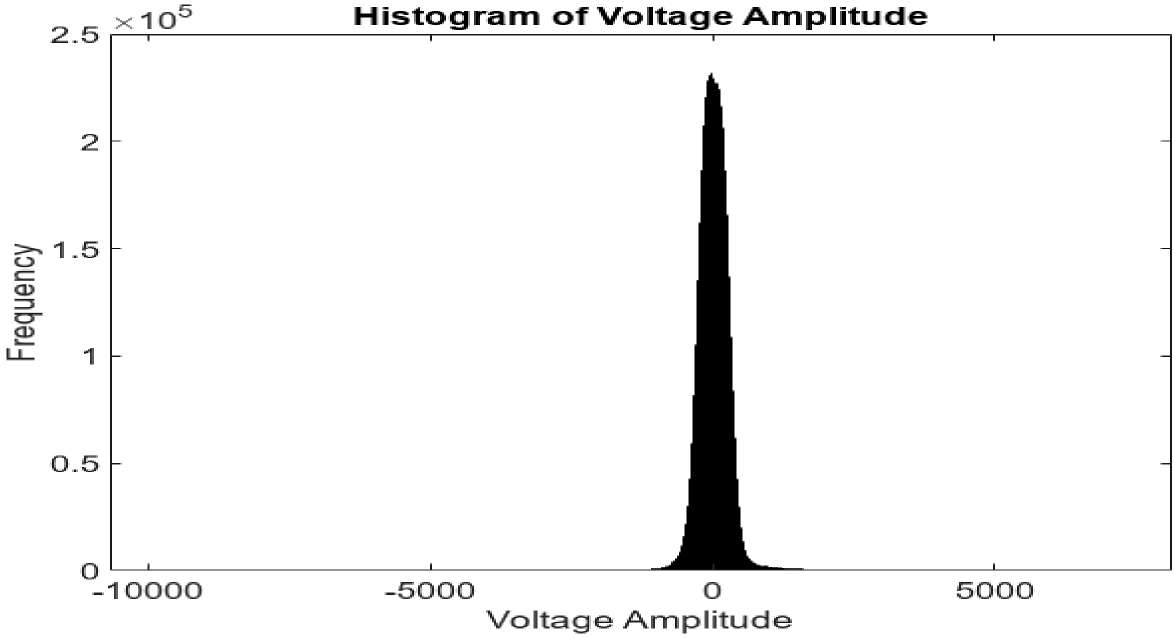


Figure 1: Raw Data Histogram

What can be inferred about the background noise by looking at this diagram (distribution, etc):

Our Voltage Amplitude seems have a normal distribution with a mean around zero, which is is expected as the voltage recorded in the neuron is very small(MV).

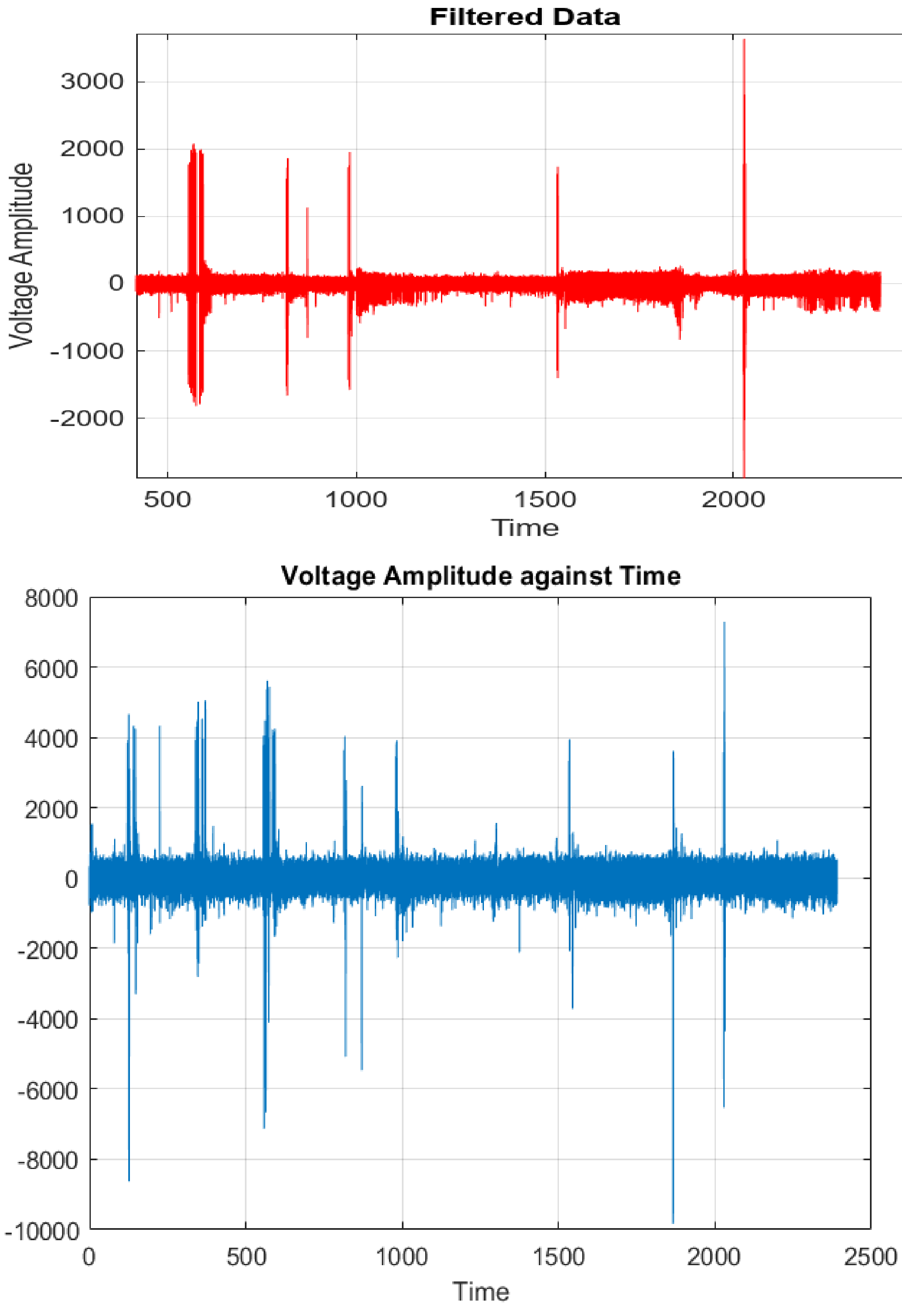


Figure 2: Raw Data(Blue) vs Filtered Data(Red)

1-2. Detecting Spikes

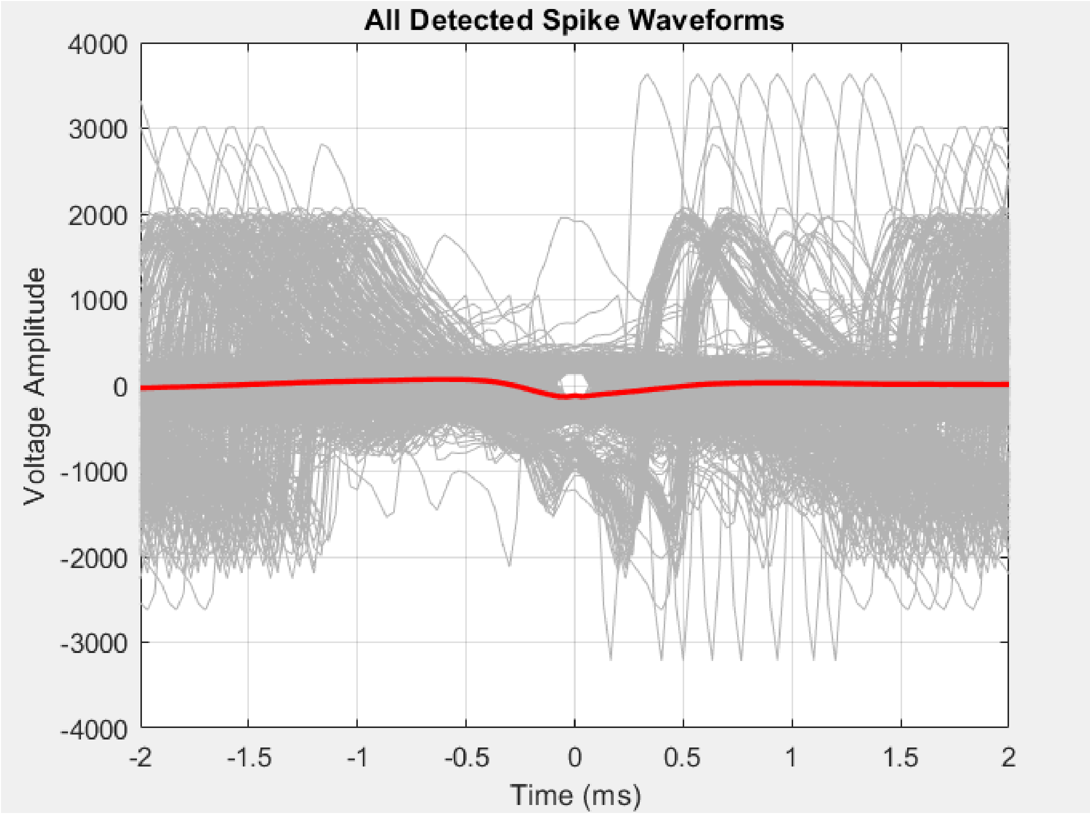


Figure 3:Waveform for Detected Spikes

Any Noticable Difference:

The voltage mean is mostly zero but has a slight curve in the peak(time 0), also showing many noise in as well.

1-3. PCA

The sum for the first Prinicpales components: 52%, which means they are capturing 52% of the features of the data.

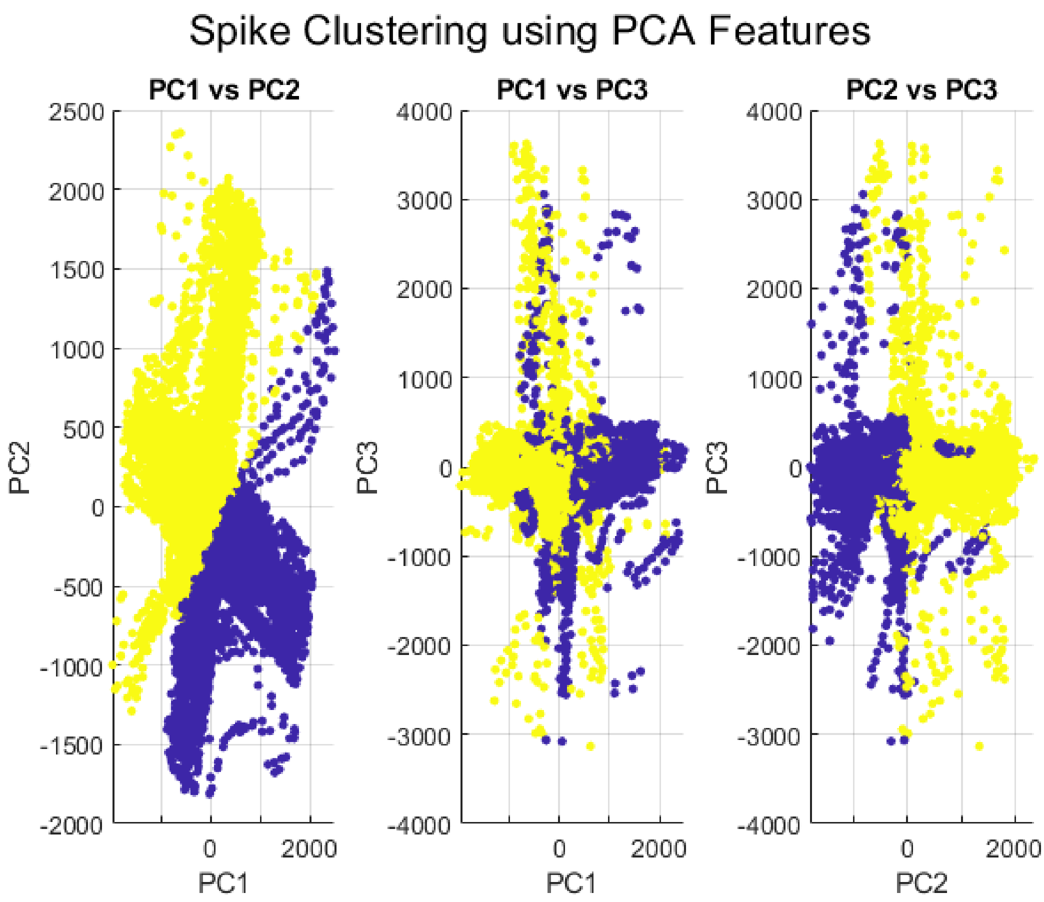


Figure 4: 2 Cluster PC's

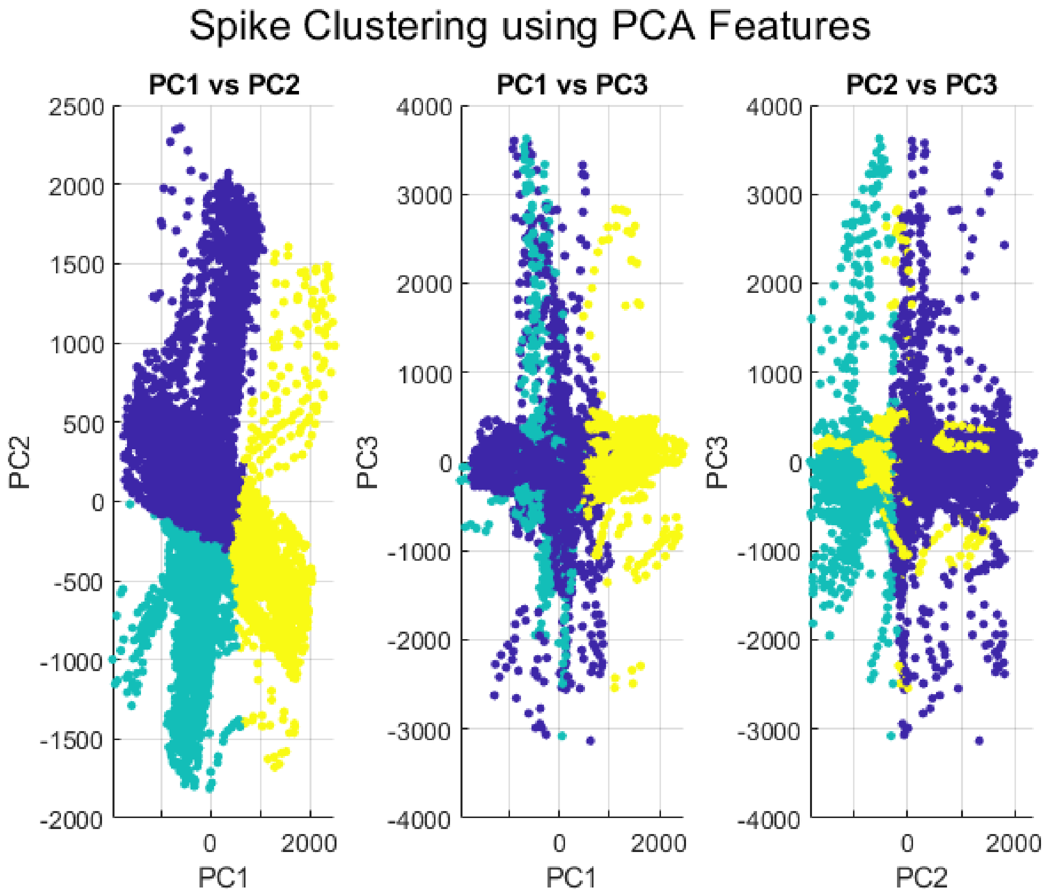


Figure 5: 3 Cluster PC's

1-4. Clustering

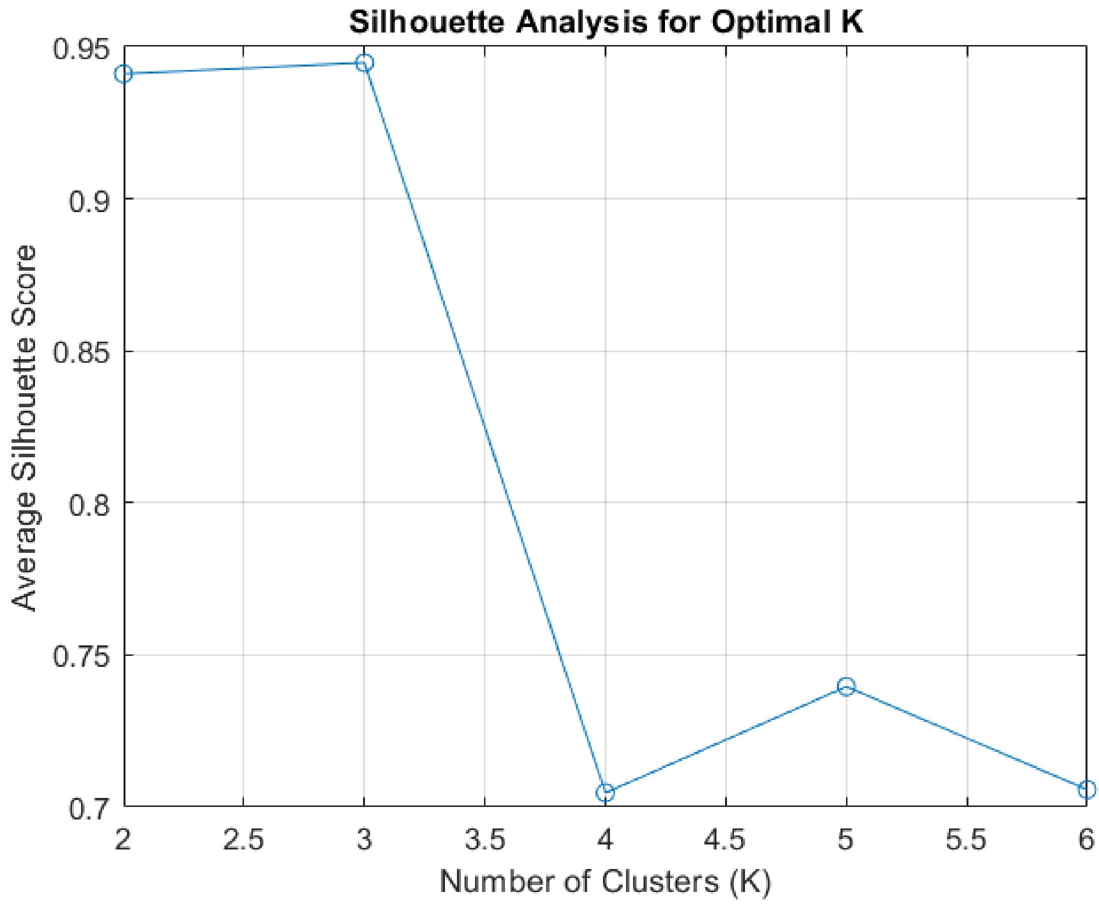


Figure 6: Clustering Silhouete Score for Different K's

Which value of K gives the best results?

Silhoute score measures how good the clustering is by measuring distance intra cluster(which should be low) and between clusters(Which should be high), here a K = 3 is showing the best and highest silhoute score, this is also the case in ross as well, this is indicating maybe we have 3 neurons or 3 types of neurons.

Comparing with Spikes.mat:

The Accuracy was zero, as the number of spikes in both case were different(Accuracies used was f1 score, recall, accuracy and etc).

Different Threshold:

This new different threshold is above 90 percent of the maximum amplitude, which is usually just noise that is continuous and not spikes, so in this section no Spikes were Detected.

TSNE:

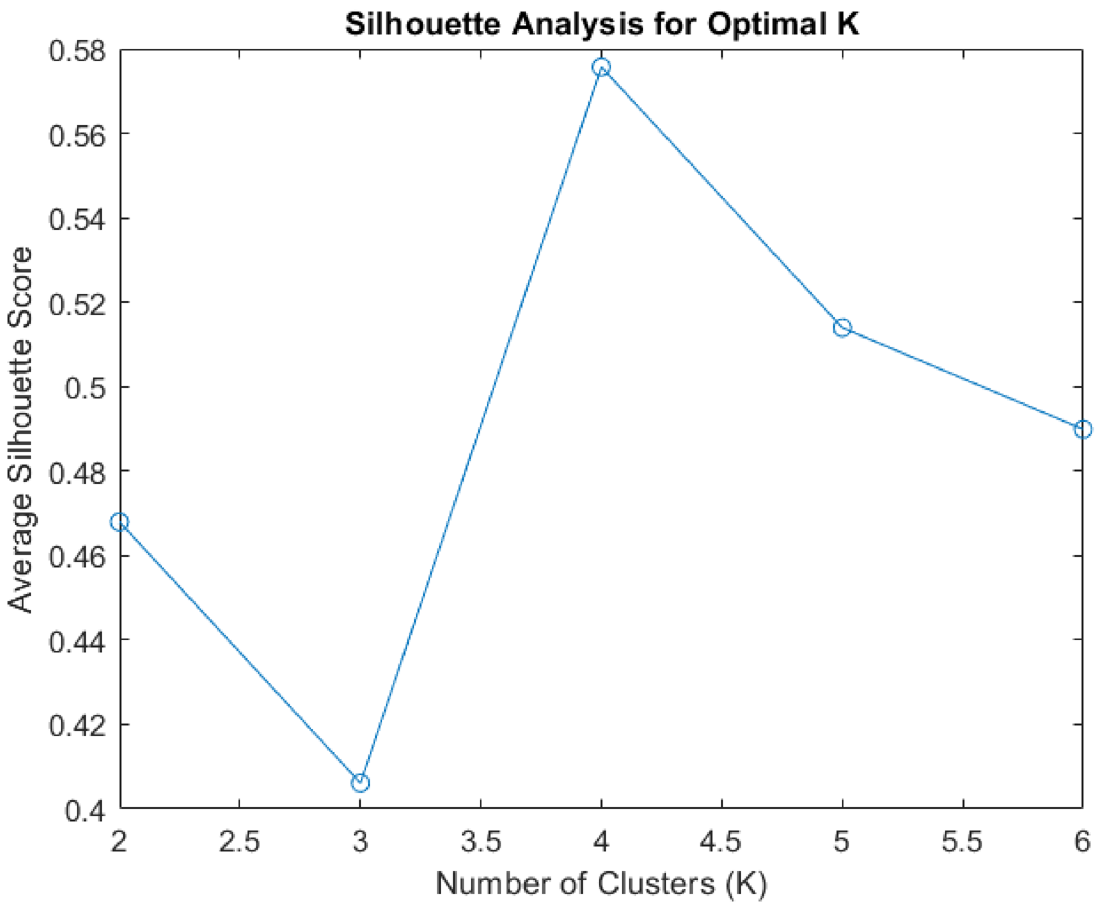


Figure 7: TSNE Silhoutee

The TSNE Silhoutee was way lower, tsne, unlike pca, does not preserve distance when projected into lower dimensions, and clustering is an alghrotim based on distance, so two points that are very distant in higher dimension, may be close in the tsne but the distance is preserved in pca.

Task 2: Sorting Using Ross

2-1. Spike Detection

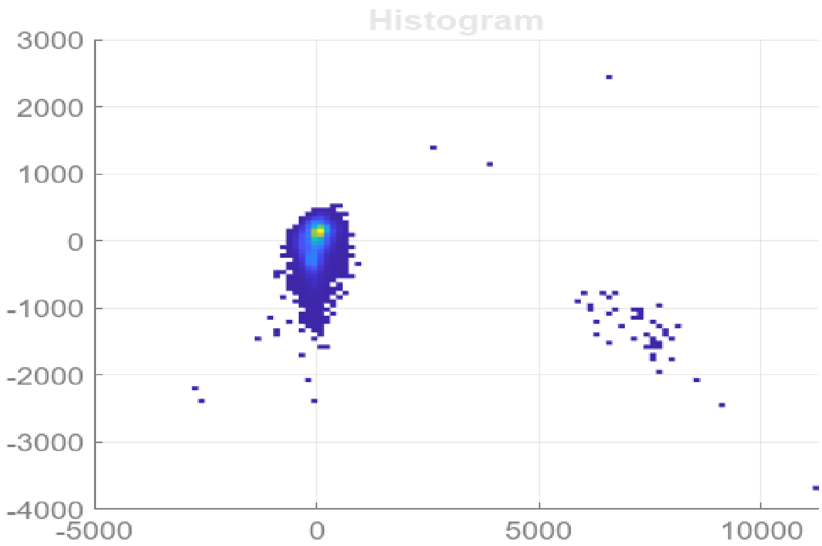


Figure 8: Spikes Detected Ross

2-2. Auto Sorting

Without Considering Noise, 3 Clusters were detected(3 neurons).

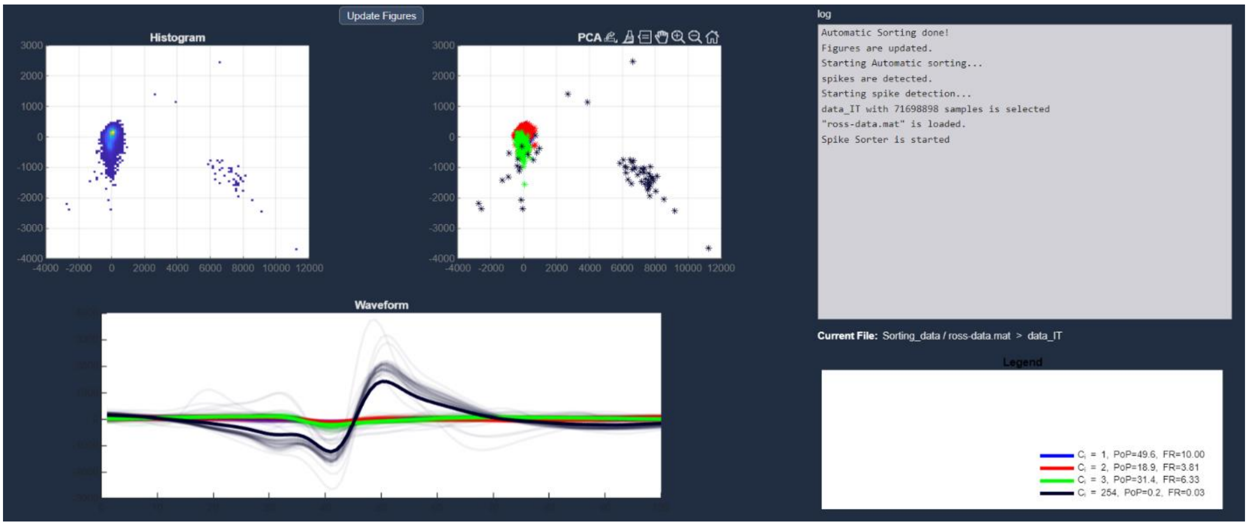


Figure 9: Auto Sorted Ross

2-3. Manual Sorting

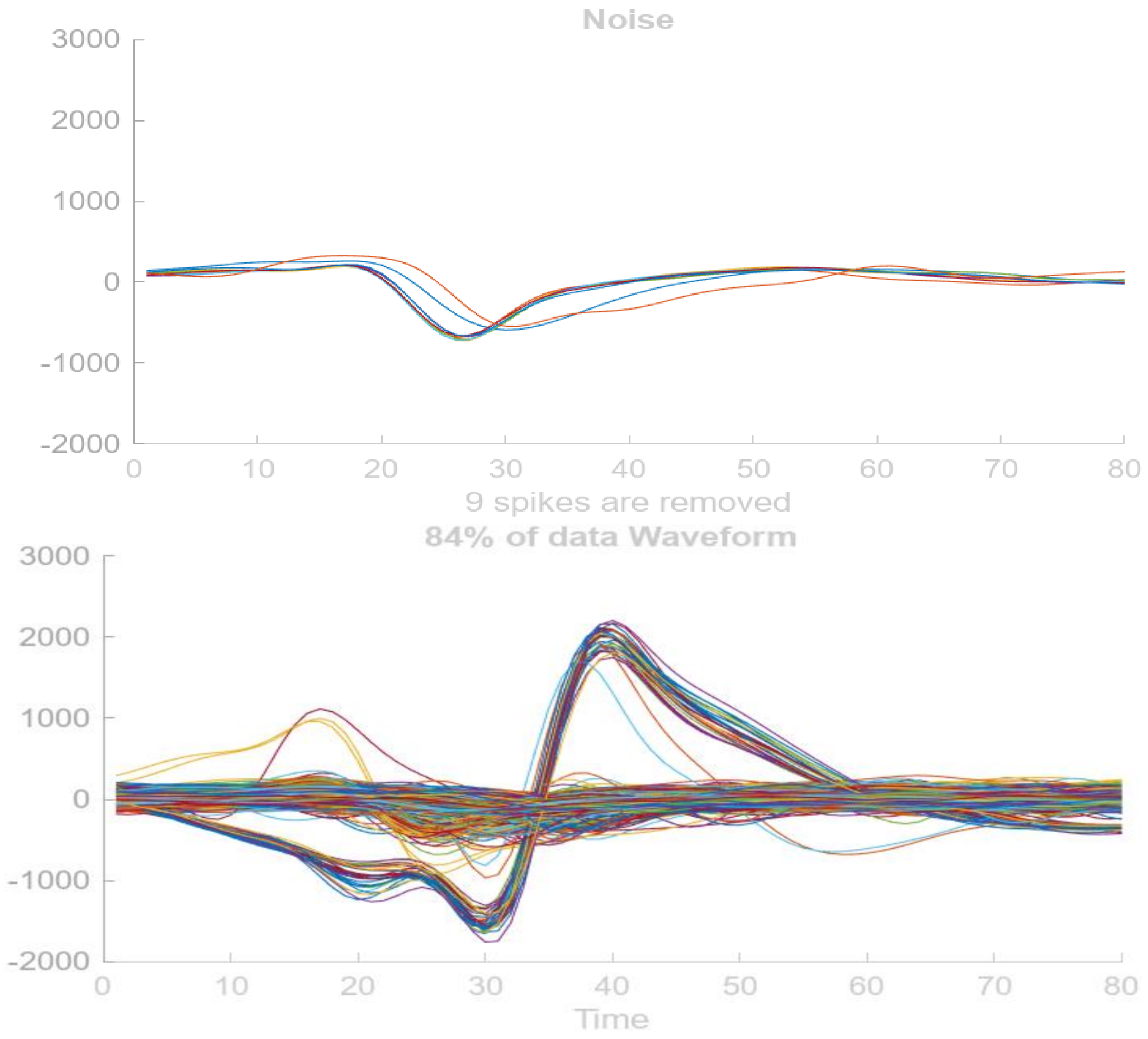


Figure 10: Denoising Spikes on Cluster 2

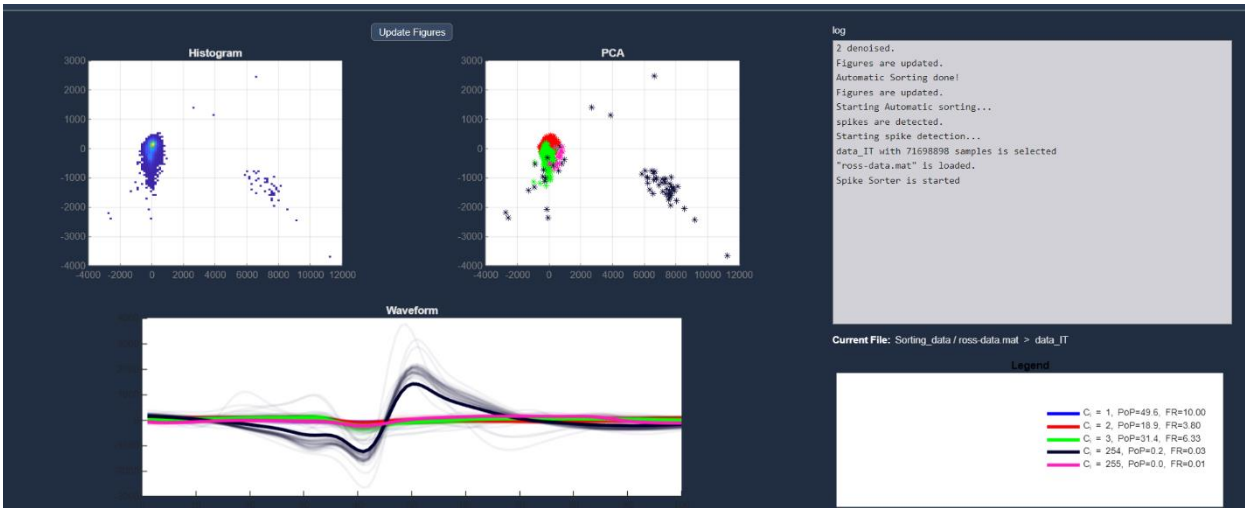


Figure 11: After removing Spikes

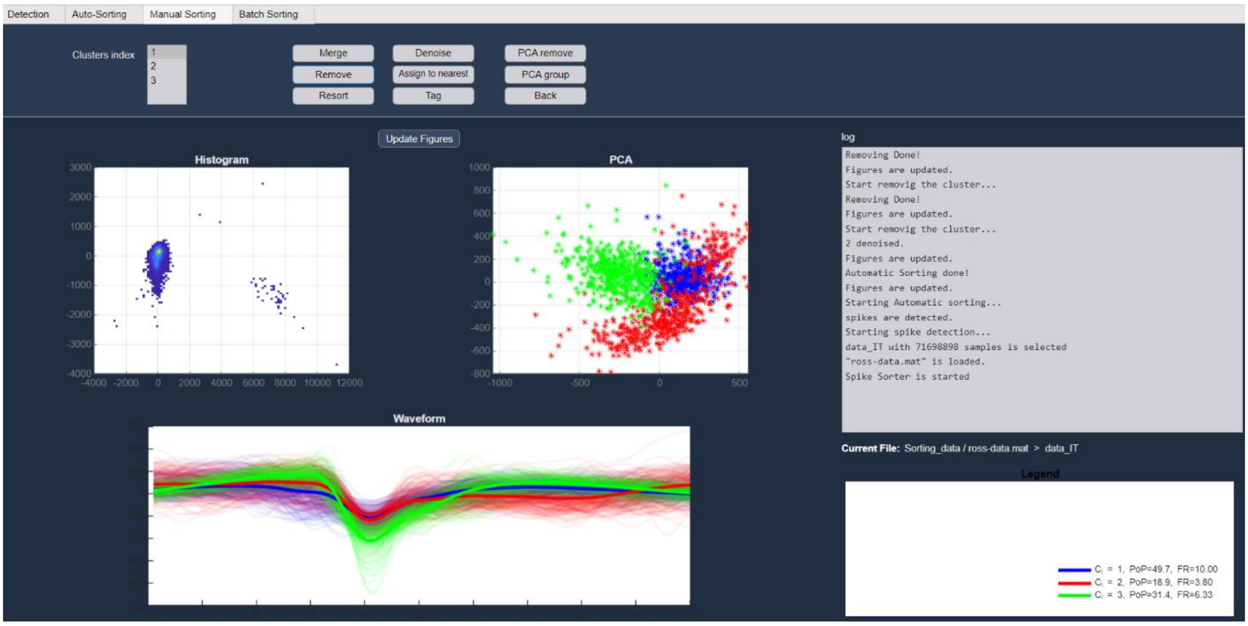


Figure 12: Removing Noisy Clusters

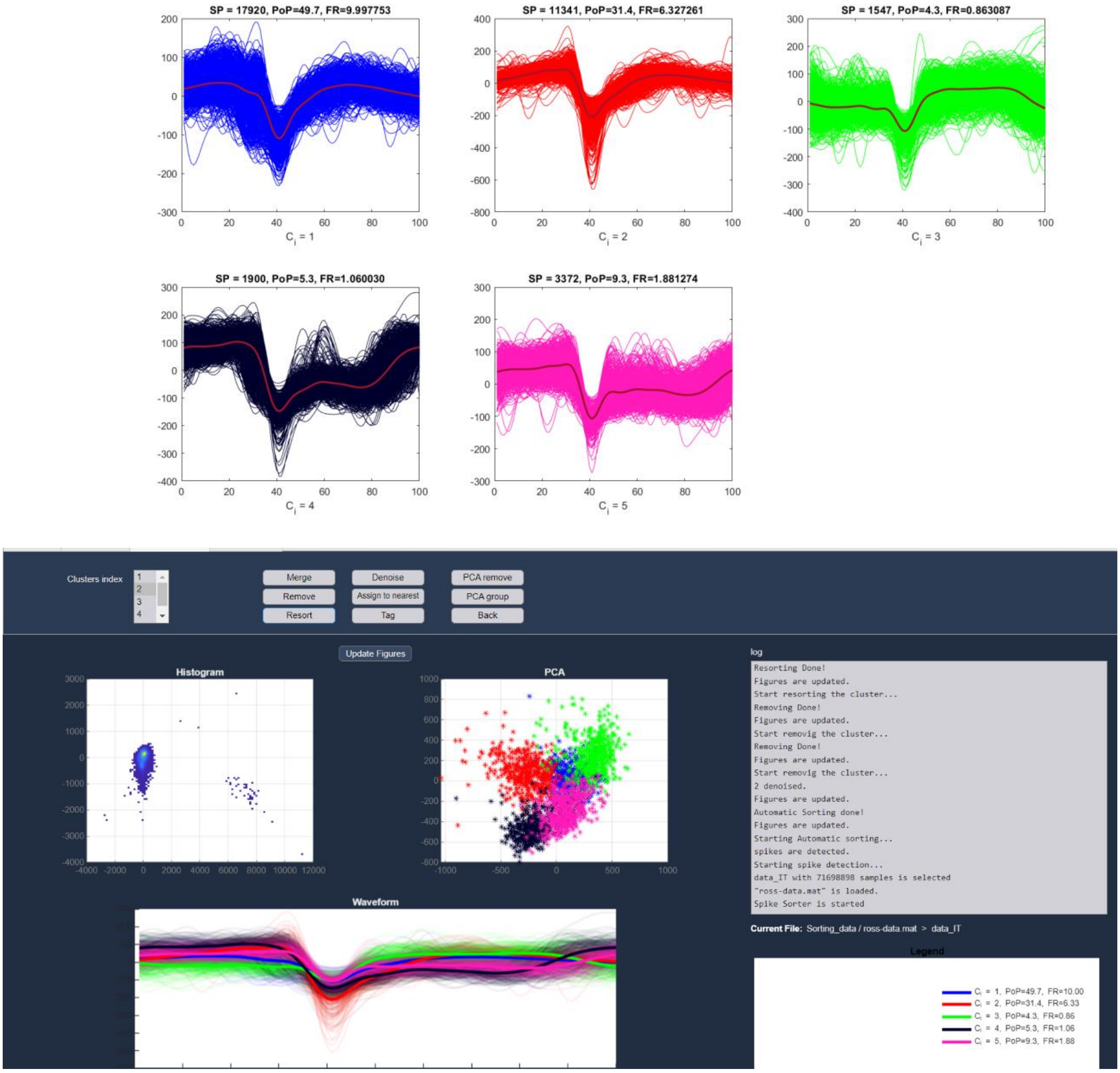


Figure 13: Re-Clustering Cluster 2

After Re-sorting Cluster 2, we got 3 clusters, which each three clusters, the FR is irrelevant and noisy, so we removed them.

All Clusters seem to be peaking in around 40.

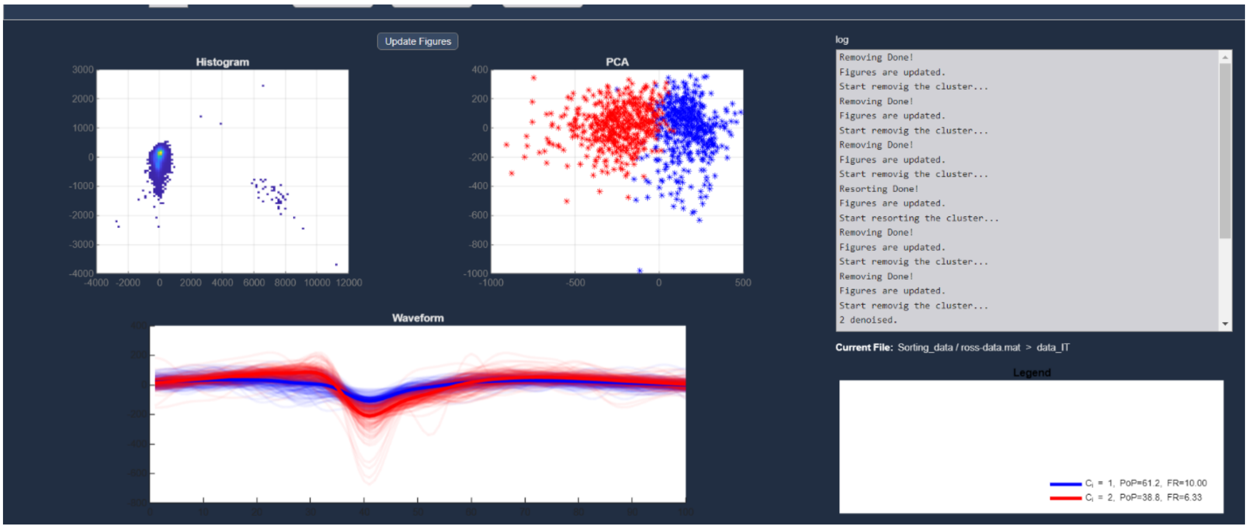


Figure 14: After Removing Noisy new Clusters.

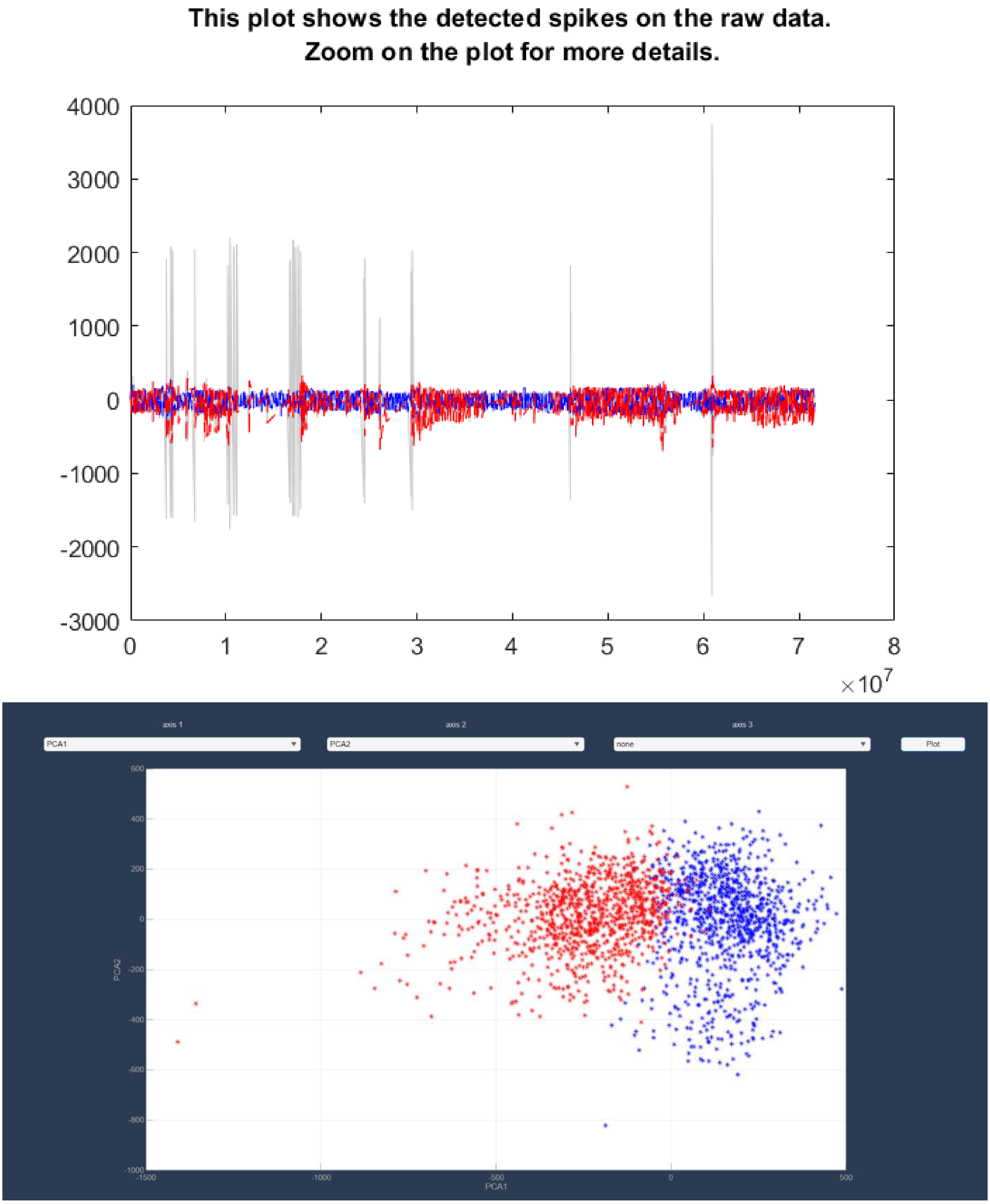


Figure 15: First Two Pca and Raw data spike's detected

Task 3: Single Neuron Analysis

3-1. PSTH

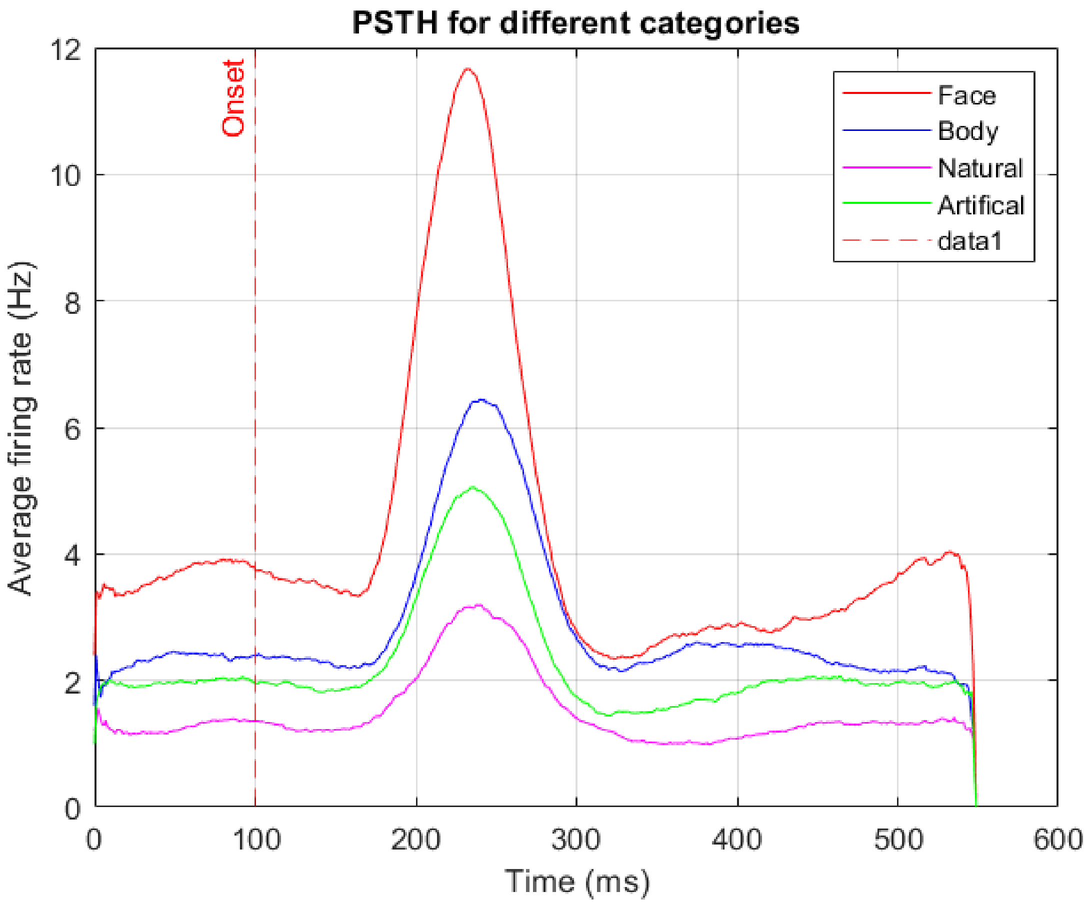


Figure 16: PSTH for different Categoris(Neuron 5)

The PSTH is looking like after around 200ms after the stimulus is shown, the neurons have fired, and they are category selective to face, meaning they are face responscie neurons because they have responed to face with higher firing rates, this increases largely in the time point where the neurons fire after the stimulus is shown.

3-2. Fano Factor

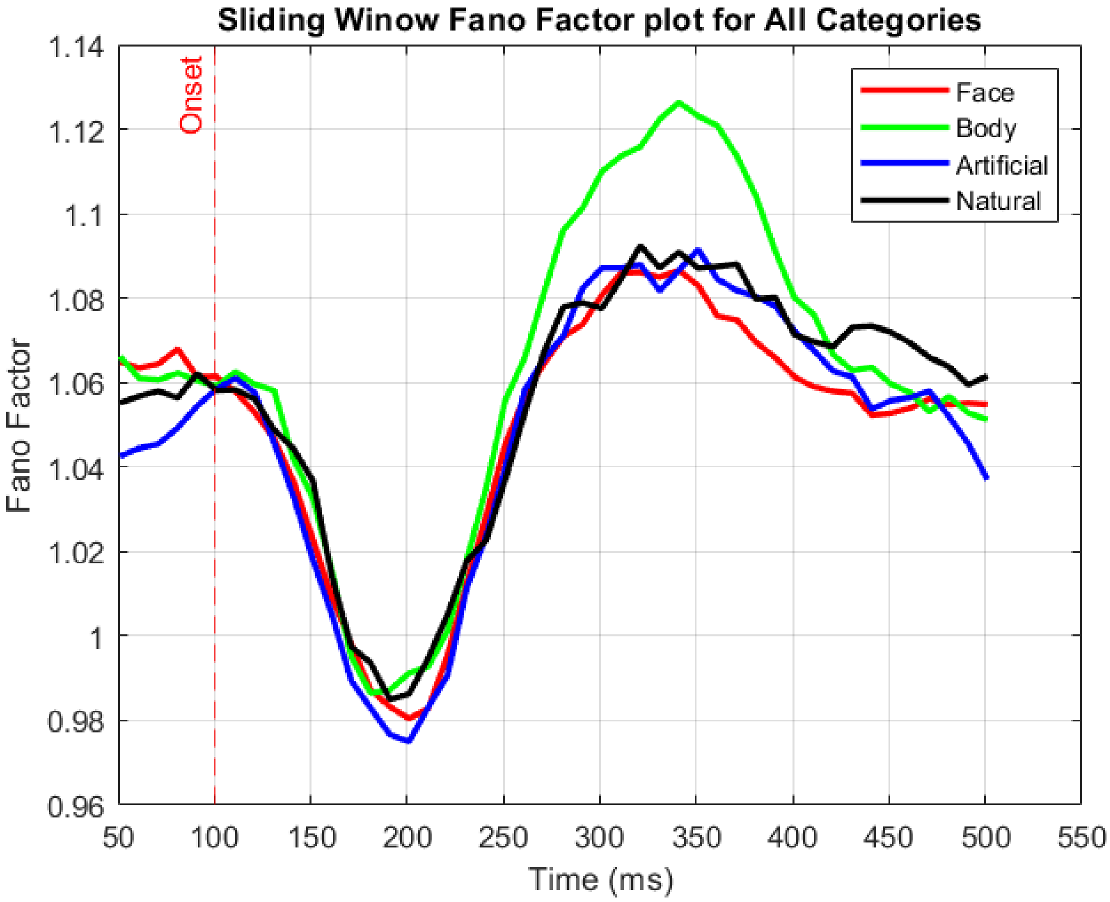


Figure 17: Regular Fano Factor

Fano Factor without mean matching showing all of them around 1.

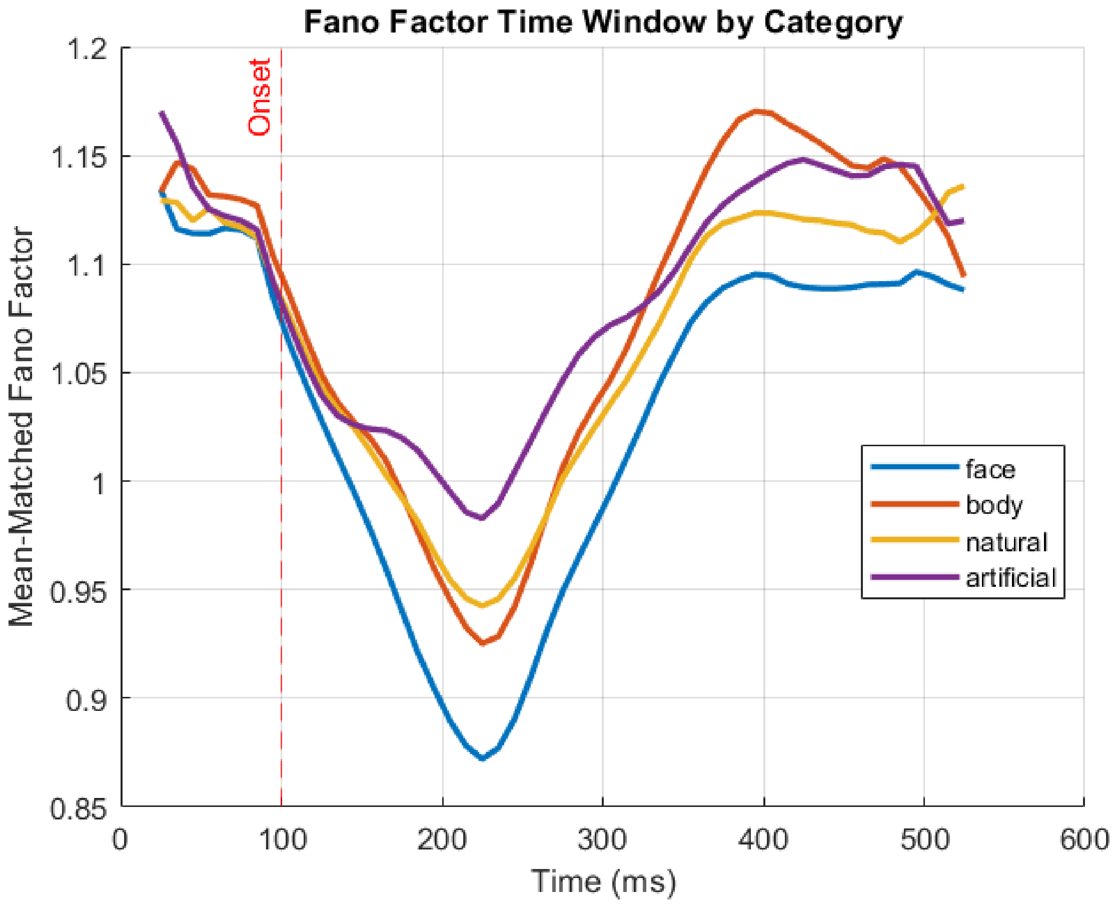
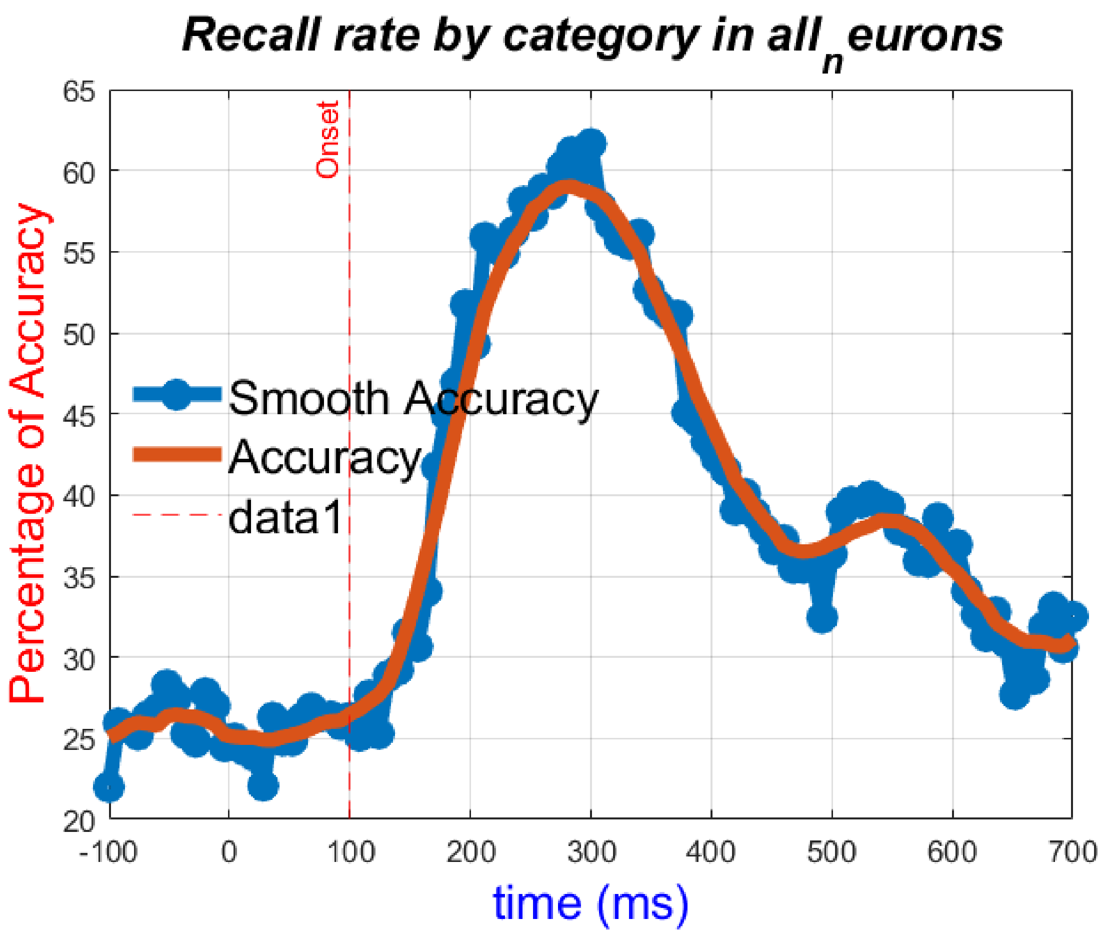


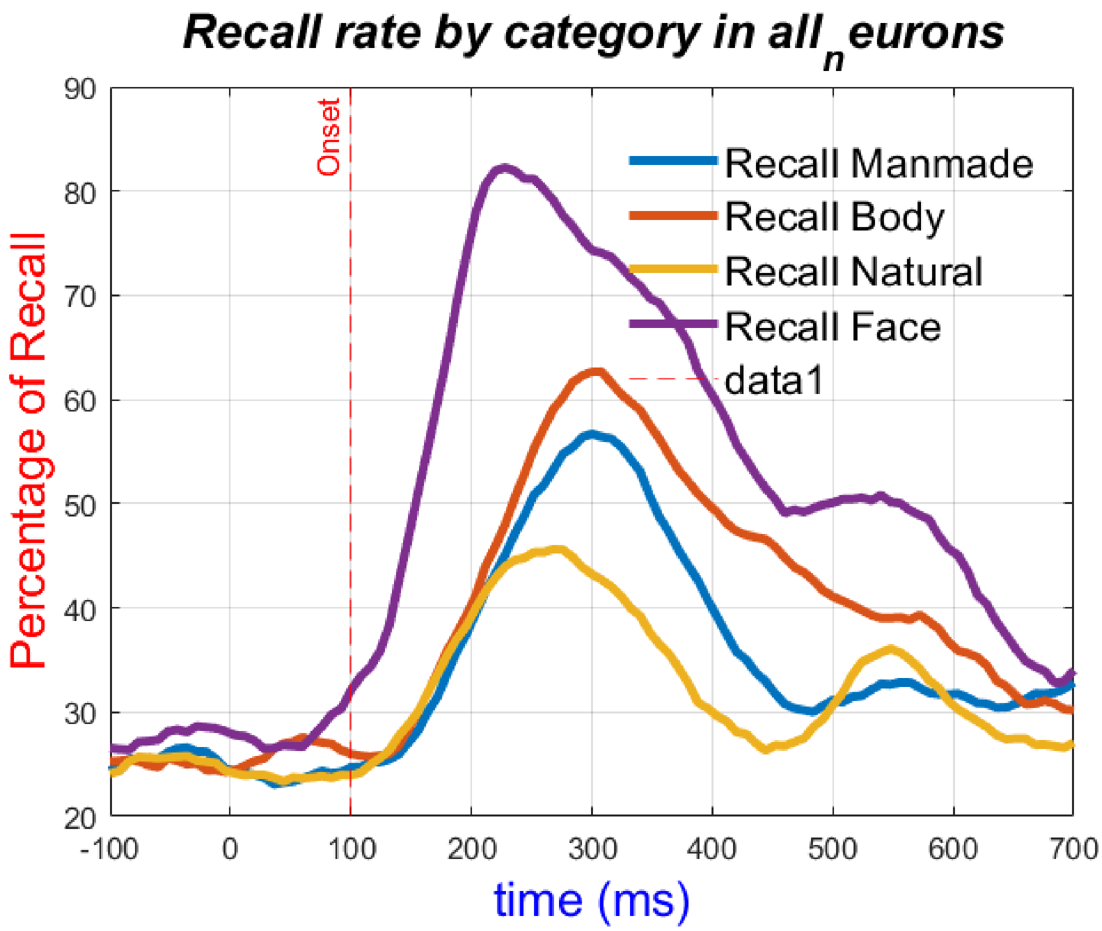
Figure 18: Mean Matched Fano Factor

Mean matched fano factor showing that face has less variability to the other categories, indicating the neurons are face selective, this variability increases in the around the same time as the neuron fires.

3-3. SVM



Recall rate by category in all_neurons
Onset
Percentage of Accuracy
Smooth Accuracy
Accuracy
data1
time (ms)
65
60
55
50
45
40
35
30
25
20
-100
0
100
200
300
400
500
600
700



Recall rate by category in all_neurons
Onset
Percentage of Recall
Recall Manmade
Recall Body
Recall Natural
Recall Face
data1
time (ms)
90
80
70
60
50
40
30
20
-100
0
100
200
300
400
500
600
700

Figure 19: SVM accuracy and recall

The svm accuracy in other time points besides the point where they fire is at the chance level, because the neurons are not firing we can't really distinguish them, but around the time when they fire the accuracy goes up to 65 percent, so when the neurons fire the svm can decode the categories.

Also the recall for face is way higher then the other ones, indicating the face category is more recognized, as the neurosn are face selective neurons. This represents that face category is represented very well compared to other because the neurons are face selective. And the worst one's are natural and manmade that is expected as they are far away from face, but body is close to it.

3-4. Time Time Decoding

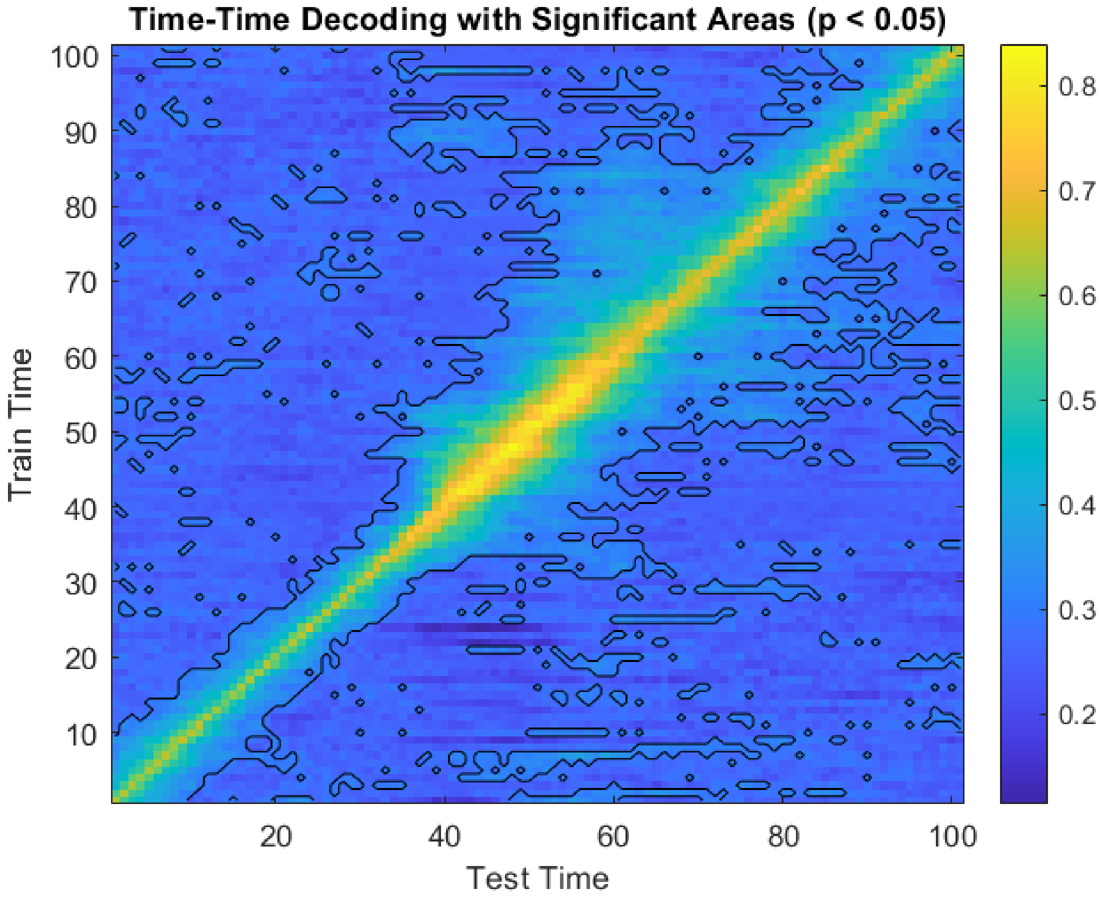


Figure 20: Time Time decoding heatmap

Arount the middle time points, where the neurons have fired it shows that it has the best decoding accuracy, again furthering our evidence that around the time the neuron fires we can decode the categories.

Also while processing on the diagonal is the highest showing feetforward processing where ddecoding evolves with ongoing time, in some parts they are signs of recurrent processing, indicating some points may use information from past points.

3-5. Mutual Information

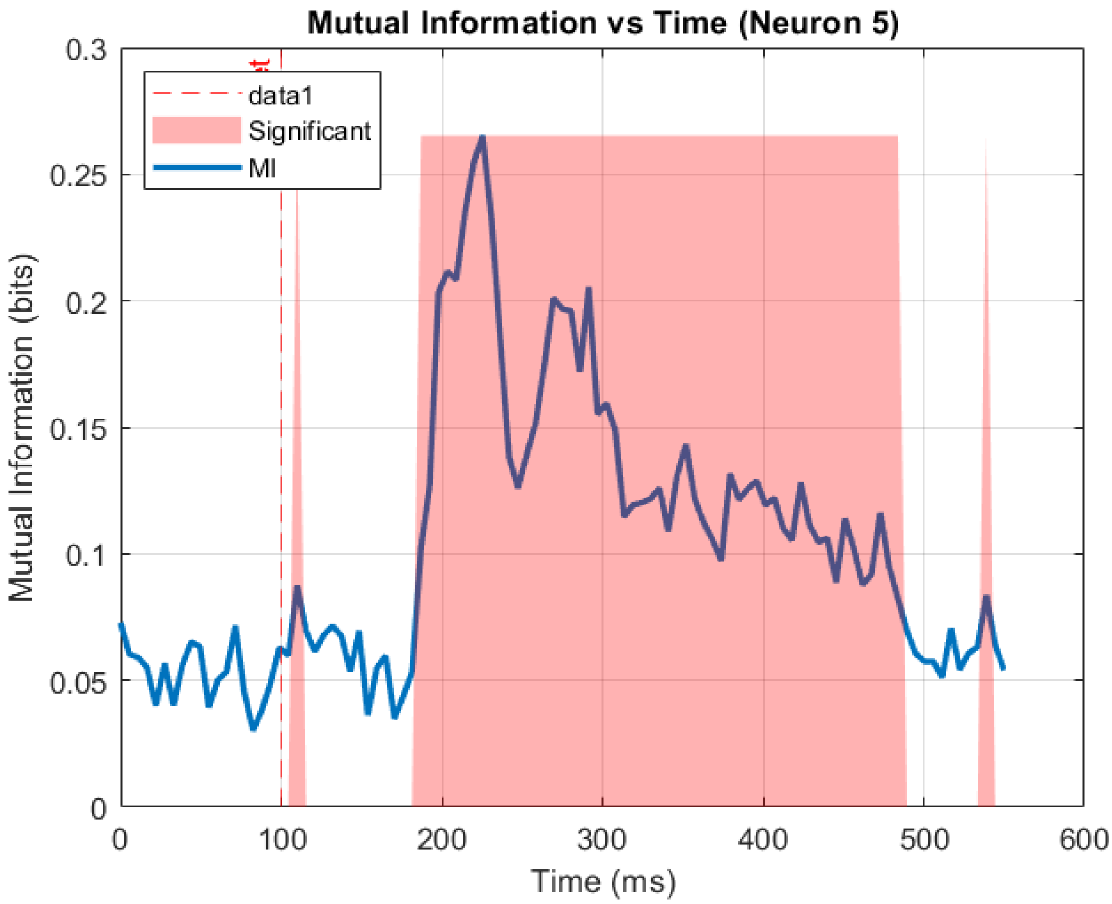


Figure 21: Mutual information plot

Again, around 150ms after the stimulus was shown, the mutual information is the highest, this is also shown in the fano factor and svm, where the highest accuracy and lowest variability and also the highest psth was in this time point. This tell us that the timing in itc after around 150-250ms it processes and encodes category-selective information, which was seen in the PNAS paper.

3-6. D Prime

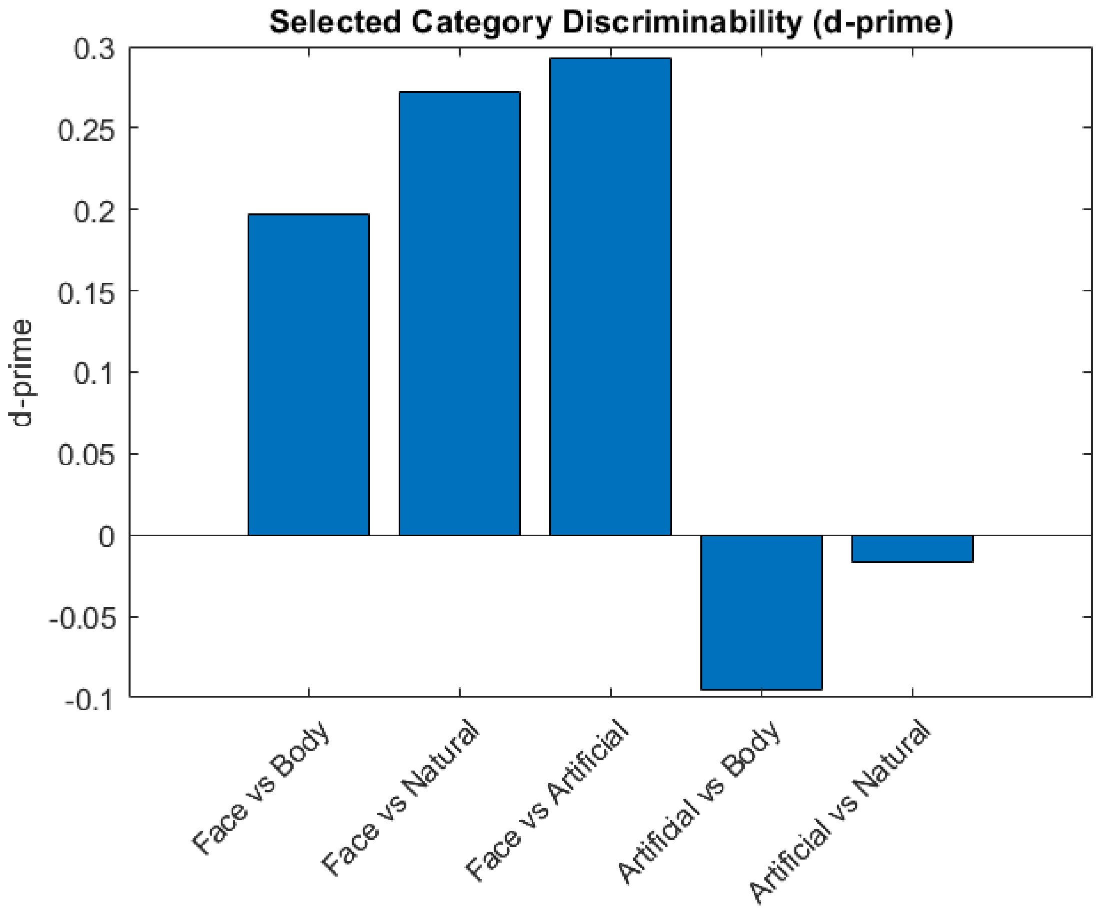


Figure 22: D-Prime for Category pais

This D prime was analyied between 150ms 250ms, in around the time that the itc processes category selective information, and it shows that the d prime for face pairs are higher as the neurons are face selective, and the lowest for like artificial and natural where the encoding for them here is low. Also category pairs when face against the lower catgoreis like natural and artificial are the best, as show in the psth they have the highest distance between them, and the lowest distance between artificial and naturals.

These also align with the svm as the svm also decodes the best in this time range.

In this neural population The neurons decode Face the most and aritifcal and natural the least.

Task 4: Analysis of Population Activity

4-1. RDM

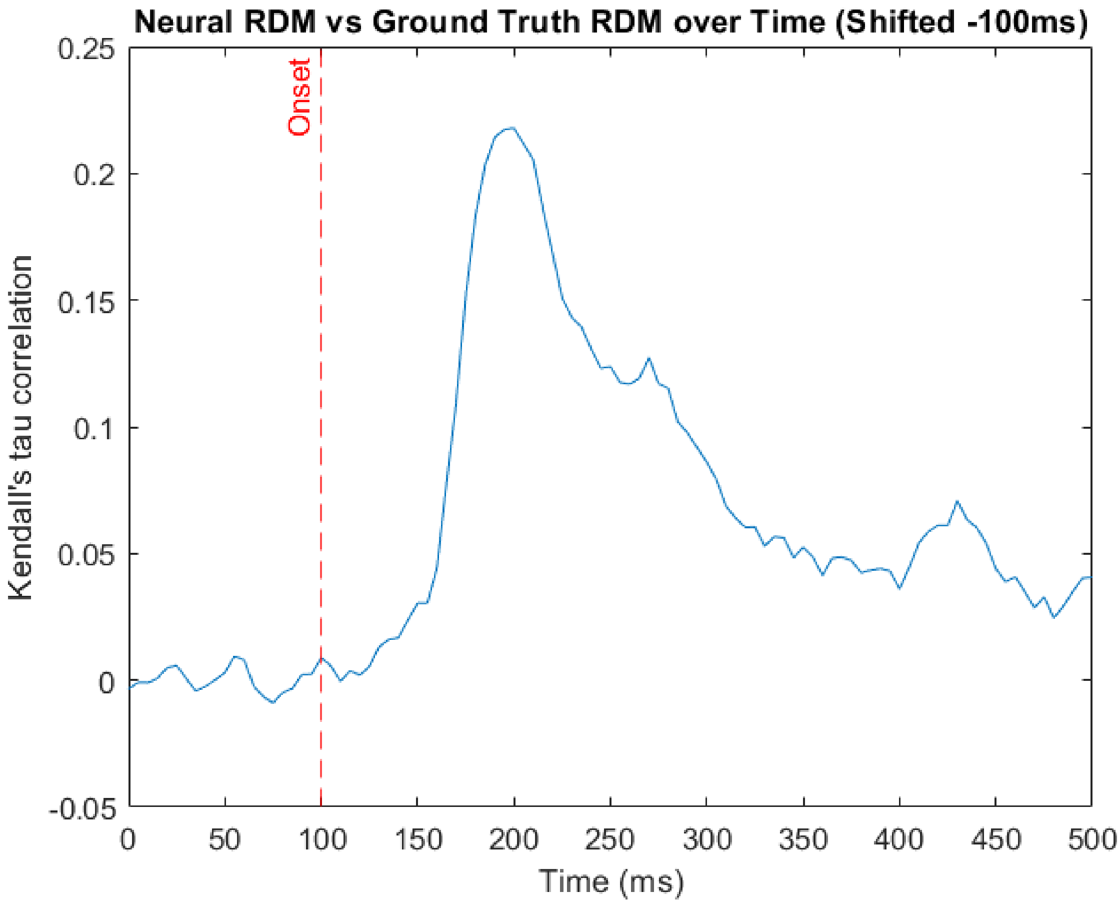
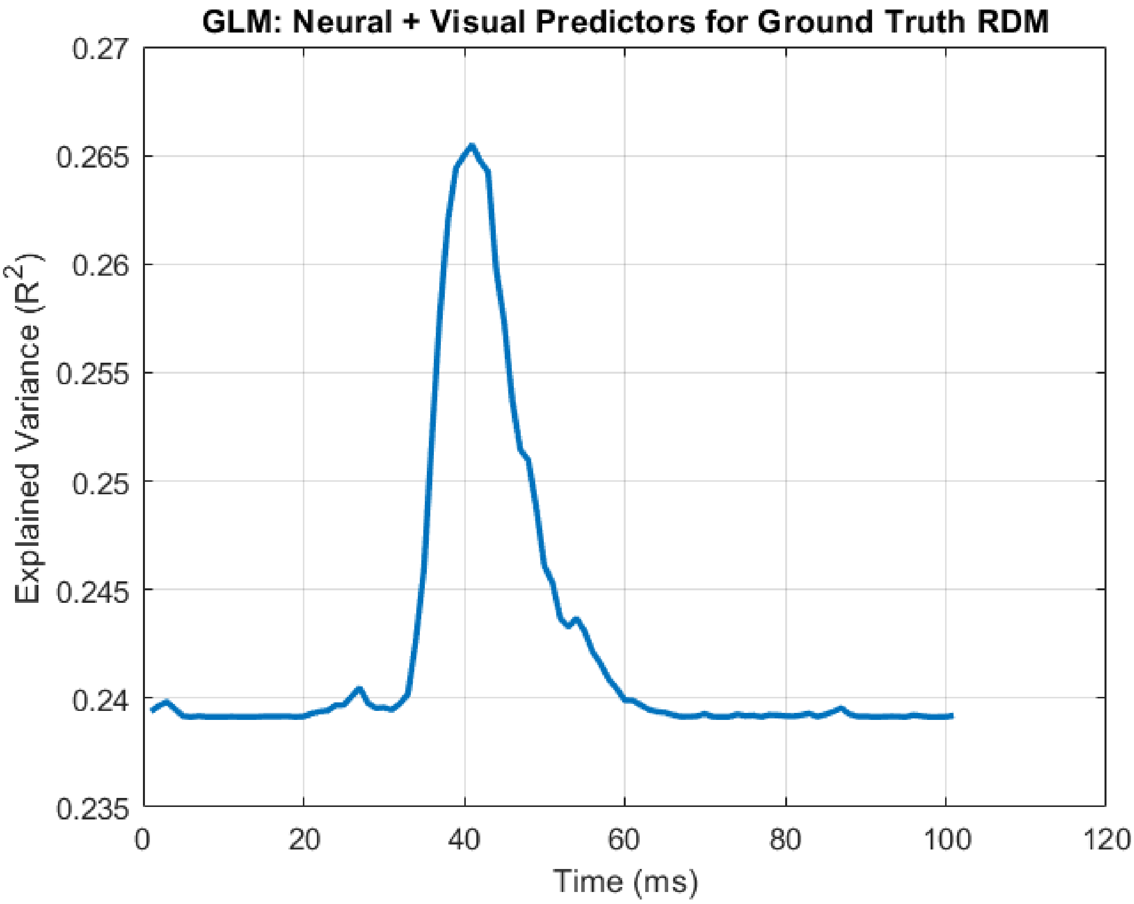


Figure 23: Rdm Correlation acorss time

Just like shown in the previous decoding tasks, around 150-200 the correlations peaks indicating in this particular time the neurons decode categories the best.

4-2. GLM


GLM: Neural + Visual Predictors for Ground Truth RDM
Explained Variance (R^2)
0.27
0.265
0.26
0.255
0.25
0.245
0.24
0.235
0
20
40
60
80
100
120
Time (ms)

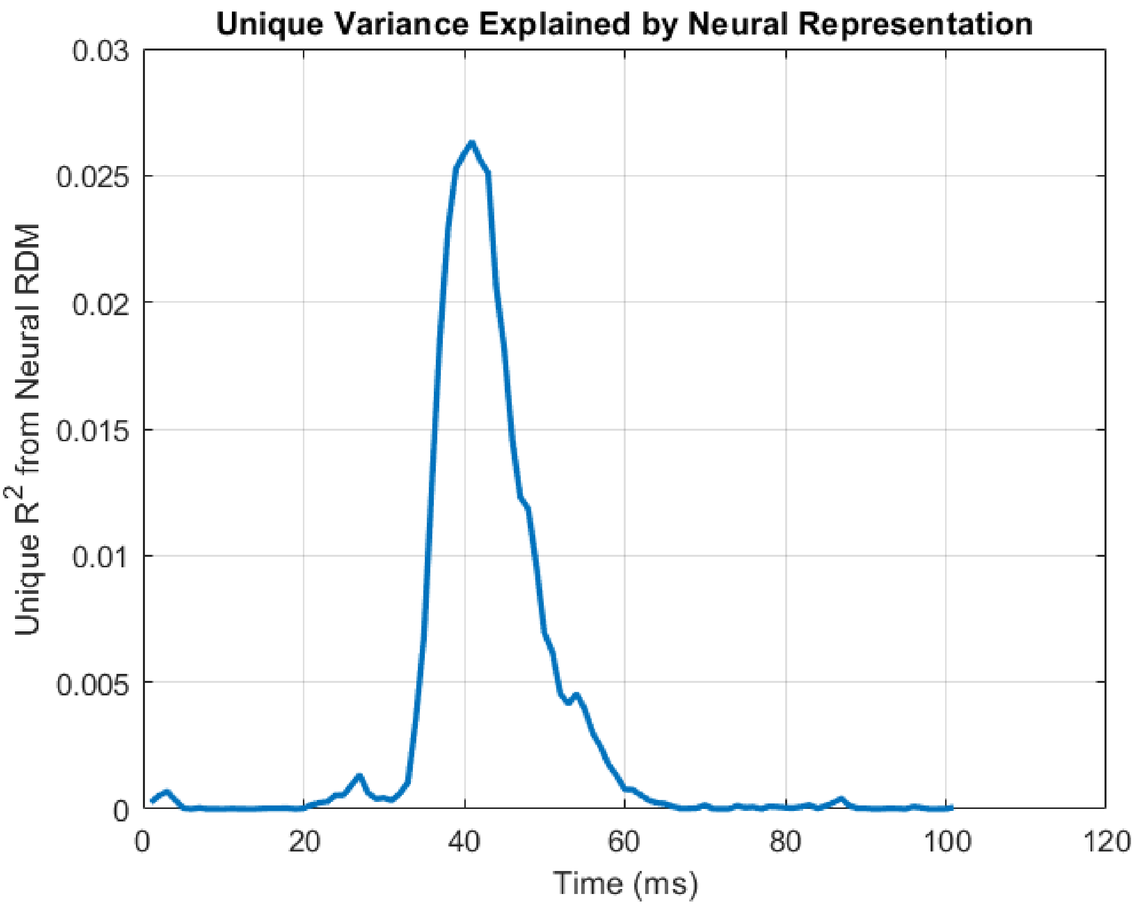

Unique Variance Explained by Neural Representation
Unique R^2 from Neural RDM
0.03
0.025
0.02
0.015
0.01
0.005
0
0
20
40
60
80
100
120
Time (ms)

Figure 24:GLM results

Yes Neural RDM does explain unique variance in the ground truth beyond what visual features can as it also peaks in around the time when the explained variance peaks.

The informations peak exactly around the time where the svm and previous decoding phases peak, suggesting more evidence where the itc after around 200ms(100-150ms after onset) decode categorical stimuli.

Task 5: PAC

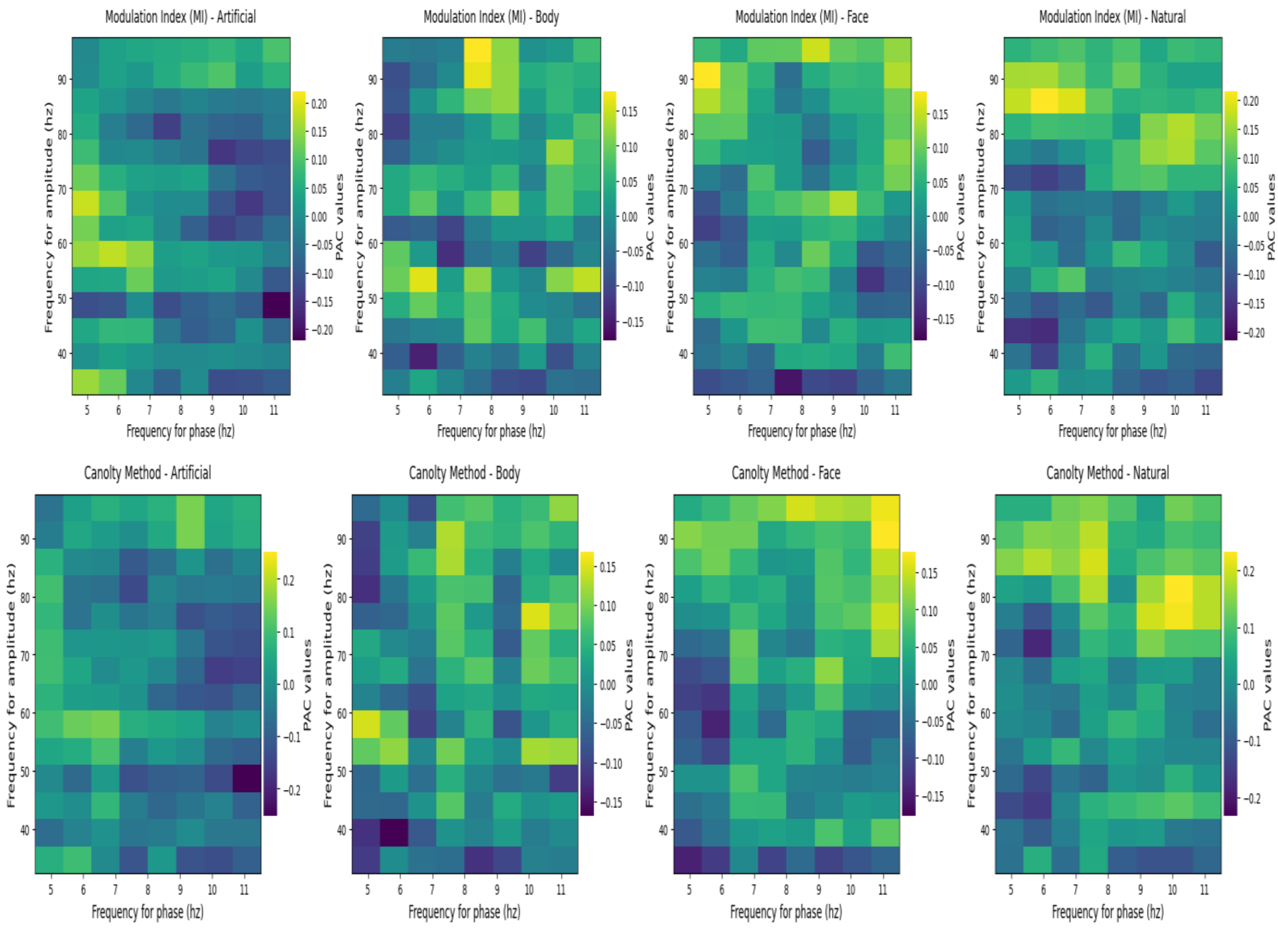


Figure 25:PAC's for differenet categories

-Both methods show coupling strength with just a little bit of difference in their focus.

Face shows more strength in phase of around 9 and ampltidue of around 60 so The phase of alpha oscillations is modulating the amplitude of gamma oscillations. While Natural shows more strernght in phase 10-11 and ampltidue 80, so in this part alpha modulates with gamma. Also artificial is strongher in phase around 5-6 and amplitude under 40, indicating modulation between thetha and low gamma. Body has strength in around 7 and 90, so high theta and high gamma.

So to sum it up:

Face: Phase around 9 Hz (alpha) modulates upper ~60 Hz (gamma) — face recognition.
Natural: Phase around 10–11 Hz (alpha) modulates upper ~80 Hz (high gamma) — scene processing.
Artificial: Phase around 5–6 Hz (theta) modulates lower ~40 Hz (low gamma) — novelty/effort.
Body: Phase around 7 Hz (high theta) modulates upper ~90 Hz (high gamma) — body/motor processing.

Face and Natural stimuli show strong alpha gamma coupling, suggesting efficient**,** familiar visual processing. Artificial stimuli rely more on theta low gamma coupling, indicating greater cognitive effort or novelty detection. Body stimuli show theta–high gamma coupling, likely reflecting sensorimotor integration. These differences suggest that the brain uses distinct frequency interactions to represent different types of sensory information based on relevance, familiarity, and processing demands.

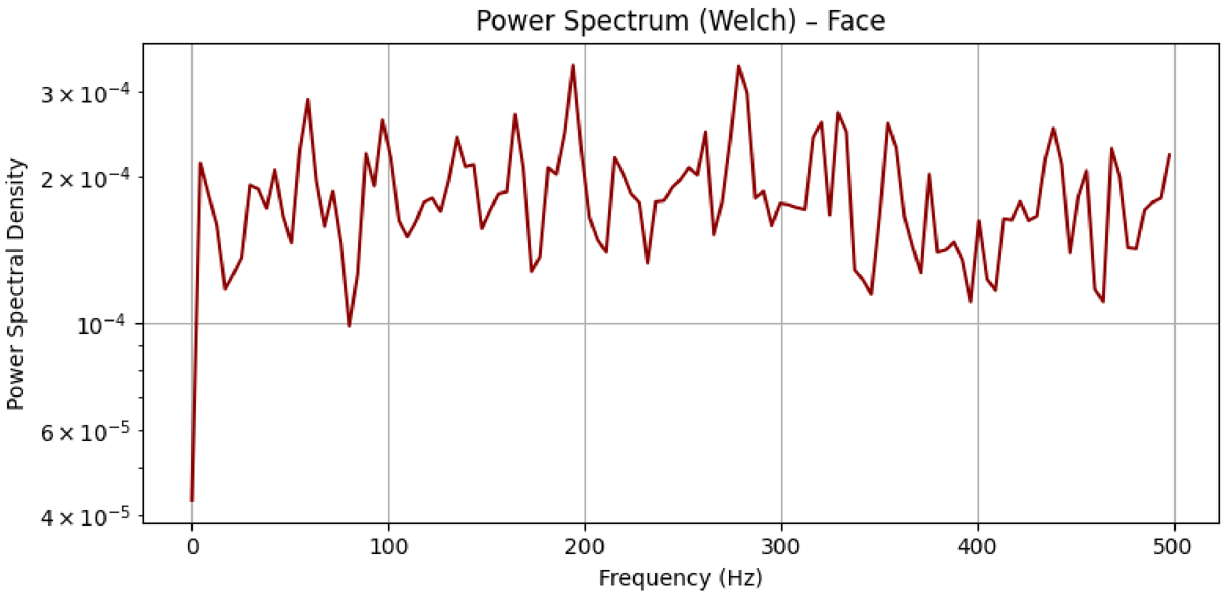


Figure 26:Power Spectrum